

questions

1) Solve the following problem with CPU scheduling algorithms. Draw a table with 4 rows with 3 columns having the following details.

Process	A_T	B_T
P_1	0	5
P_2	0	4
P_3	0	3

There is a context switch in the CPU. The context switch time is b/w processes of 5 seconds.

Apply the following CPU scheduling algorithms:-

* Preemptive algorithms:-

i) SRTF

(ii) Priority scheduling.

* Non-preemptive algorithms:-

i) FCFS

ii) SJF

iii) Priority scheduling.

Solve each algorithm with its Gantt chart, CT, TAT, WT, average TAT and average WT.

Q1. Given,

Process	A_T	B_T
P_1	0	5
P_2	0	4
P_3	0	3

→ Context switch Time = 5 sec

formulas:-

$$TAT = CT - AT$$

$$WT = TAT - BT$$

(i) FCFS - (first come first serve)

Gantt chart.

$|P_1|CS|P_2|CS|P_3|$
 0 5 10 14 19 22

Process	A_T	B_T	CT	$TAT = CT - AT$	$WT = TAT - BT$
P_1	0	5	5	5	0
P_2	0	4	14	14	10
P_3	0	3	22	22	19

$$\text{Average } WT = \frac{0+10+19}{3} = \frac{19}{3} = 9.67$$

$$\text{Average } TAT = \frac{5+14+22}{3} = 13.67$$

ii) SJF (shortest Job first)

Since SJF selects the process with the shortest BT:-

$P_3(3) \rightarrow P_2(4) \rightarrow P_1(5)$

order:- $P_3 \rightarrow P_2 \rightarrow P_1$

Gantt chart:-

$$\begin{array}{c}
 0 \quad 3 \quad 8 \quad 12 \quad 17 \quad 22 \\
 |P_3|C_3|P_2|C_2|P_1|
 \end{array}$$

Calculation:-

Process	AT	BT	CT	TAT	WT
P_1	0	5	22	22	17
P_2	0	4	12	12	8
P_3	0	3	3	3	0

$$\text{Average WT} = \frac{17+8+0}{3} = 8.33$$

$$\text{Average TAT} = \frac{22+12+3}{3} = 12.33$$

iii) SRTF :-

All process arrive at 0, so no preemption occurs.

order : $P_3 \rightarrow P_2 \rightarrow P_1$

Gantt chart:-

$$\begin{array}{c}
 |P_3|C_3|P_2|C_2|P_1| \\
 0 \quad 3 \quad 8 \quad 12 \quad 17 \quad 22
 \end{array}$$

- ii) Draw the execution timeline for SRTF.
- iii) Explain why the selected algorithm improves user experience.
- (iv) which task receives the fastest response time?

Ans:

i) SRTF is the suitable scheduling algorithm because SRTF always selects the task with the shortest remaining CPU burst time.

ii) SRTF execution timeline.

* In SRTF, the process with the shortest remaining burst time is selected.

* 0-3: MP runs.

$$\text{Remaining MP} = 20 - 3 = 17$$

* At 3: TE arrives with $BT = 1$

TE is shorter than MP $\rightarrow$ TE runs from 3-4

* 4-5: MP continues

$$\text{Remaining MP} = 16$$

* At 5: WN arrives with $BT = 2$

WN is shorter than MP $\rightarrow$ WN runs

* 6:- CL arrives with $BT = 4$, but WN has only 1 unit remaining, so WN continues.

* 5-7: WN completes

* 7-11: compare $MP = 16$ and $CL = 4 \rightarrow$ CL runs

* 11-27: CL finishes, so remaining MP runs until completion.

Process	A _T	B _T	C _T	TAT	W _T
P ₁	0	5	22	22	17
P ₂	0	4	12	12	8
P ₃	0	3	3	3	0

$$\text{Average } W_T = \frac{17+8+0}{3}$$

$$= 8.33$$

$$\text{Average TAT} = \frac{22+12+3}{3} = 12.33$$

iv) Priority scheduling:

cannot solve because priority values are not given.

2) A mobile operating system has the following tasks.

Task	A _T	CPU burst
Music players(mp)	0	20
Touch event(TE)	3	1
What'sapp Notification(WN)	5	2
Camera launch(CL)	6	4

i) Which scheduling algorithm is most suitable: FCFS, SJF, SRTF (or) RR?

3) Define the following terms:-

i) starvation

ii) convoy effect

iii) context switching

iv) aging:-

i) starvation:-

starvation occurs when a process waits for a very long time because other processes are repeatedly selected for execution.

eg:- A low priority process keeps waiting while high priority process continuously execute

(ii) convoy effect:-

convoy effect occurs in FCFS scheduling when a long process makes many short processes wait behind it

eg:- A long process runs first, causing several short processes to wait

iii) context switching:-

It is the process of saving the state of the currently running process and loading the state of another process so that the CPU can switch between them.

eg:- The OS saves process P_1 's state and loads P_2 's state.

R_7
0
0
0
1

ii) Aging:-

Aging is a technique used to prevent starvation by gradually increasing the priority of the process that has been waiting for a long time.

Eg:- A low-priority process gets a higher priority as its waiting time increases.

4) Consider the following process:-

Process	AT	BT	Priority
P ₁	0	4	3
P ₂	0	9	1
P ₃	0	6	2
P ₄	0	4	3
P ₅	0	9	1
P ₆	0	6	2
P ₇	1	4	3
P ₈	1	9	1
P ₉	2	6	2

Apply the CPU scheduling algorithms to given processes

i) FCFS:-

FCFS means first come first serve.

Table:-

Process
P ₁
P ₂
P ₃
P ₄
P ₅
P ₆
P ₇
P ₈
P ₉
Average

(ii) SJF

→ Gantt chart:-

mp | TE | mp | wn | CL | mp
0 3 4 5 7 11 27

final execution order:-

mp → TE → mp → wn → CL → mp

(iii) SRTF gives quick response to short tasks such as touch events, notifications and camera launch.

∴ mobile devices feels.

* fast

* Responsive

* Interactive

→ SRTF improves responsiveness by prioritizing shorter remaining tasks.

iv) Response Time

→ Response time = first cpu start time - A_i

Task	A_i	B_i	first cpu start	CT	TAT ($CT - A_i$)	W_i^0 ($TAT - B_i$)	R_i
mp	0	20	0	27	$27 - 0 = 27$	7	0
TE	3	1	3	4	$4 - 3 = 1$	0	0
wn	5	2	5	7	$7 - 5 = 2$	0	0
CL	6	4	7	11	$11 - 7 = 4$	1	1

→ mp, TE and wn all have $R_i = 0$, so music player, touch event and whatsapp notification share the fastest response time of 0 seconds.

Gantt chart:

| P₁ | P₂ | P₃ | P₄ | P₅ | P₆ | P₇ | P₈ | P₉ |
 0 4 13 19 23 32 38 42 51 57

Table:-

Process	AT	BT	CT	TAT	WT	RT
P ₁	0	4	4	4	0	0
P ₂	0	9	13	13	4	4
P ₃	0	6	19	19	13	13
P ₄	0	4	23	23	19	19
P ₅	0	9	32	32	23	23
P ₆	0	6	38	38	32	32
P ₇	1	4	42	41	37	37
P ₈	1	9	51	50	41	41
P ₉	2	6	57	55	47	47

Average:-

Average TAT = 30.56

Average WT = 24.22

(ii) SJF :- SJF means shortest job first. Since it is non-preemptive, once a process starts, it finishes completely.

order:-

shortest available processes are selected first.

P₁ → P₄ → P₇ → P₃ → P₆ → P₅ → P₂ → P₈ → P₉

* 18-24 $\rightarrow P_6$

* 24-30 $\rightarrow P_9$

* 30-39 $\rightarrow P_2$

* 39-48 $\rightarrow P_5$

* 48-52 $\rightarrow P_8$

Gantt chart

0 4 8 12 18 24 30 39 48 52
| P₁ | P₄ | P₇ | P₃ | P₆ | P₉ | P₂ | P₅ | P₈ |

Process	CT	TAT	WT	RT
P ₁	4	4	0	0
P ₄	8	8	4	4
P ₇	12	11	7	7
P ₃	18	18	12	12
P ₆	24	24	18	18
P ₉	30	28	22	22
P ₂	39	39	30	30
P ₅	48	48	39	39
P ₈	52	56	47	47

Average TAT = 26.22

Average WT = 19.89

V) Priority scheduling

Priority 1 = highest priority.

At time 0, P_2 and P_5 have the highest priority, so they are selected first.

→ Average TAT = 36.22

→ Average WT = 29.89

order:-

$P_2 \rightarrow P_5 \rightarrow P_8 \rightarrow P_3 \rightarrow P_6 \rightarrow P_9 \rightarrow P_1 \rightarrow P_4 \rightarrow P_7$

Gantt chart:-

0 9 18 27 33 39 45 49 53 57
 $|P_2|P_5|P_8|P_3|P_6|P_9|P_1|P_4|P_7|$

Process	AT	BT	CT	TAT	WT	RT
P_2	0	9	9	9	0	0
P_5	0	9	12	18	9	9
P_8	1	9	27	26	17	17
P_3	0	6	33	33	27	27
P_6	0	6	39	39	33	33
P_9	2	6	45	43	37	37
P_1	0	4	49	49	45	45
P_4	0	4	53	53	49	49
P_7	1	4	57	56	52	52

Gantt chart:-

0 4 8 12 18 24 30 35 40 52
 | P₁ | P₄ | P₇ | P₃ | P₆ | P₉ | P₂ | P₅ | P₈ |

Table

Process	A _T	B _T	C _T	TAT	WT	A _T
P ₁	0	4	4	4	0	0
P ₄	0	4	8	8	4	4
P ₇	1	4	12	11	7	7
P ₃	0	6	18	18	12	12
P ₆	0	6	24	24	18	18
P ₉	2	6	30	28	22	22
P ₂	0	9	39	39	30	30
P ₅	0	9	48	48	39	39
P ₈	1	9	52	56	42	42

Average TAT = 26.22

Average WT = 19.89

iii) SRTF.

- SRTF means shortest remaining time first
- It is the preemptive version of SJF

Execution:

- * 0-4 → P₁
- * 4-8 → P₄
- * 8-12 → P₇
- * 12-18 → P₃